The Journal of Open Source Software

SemanticDistance: An R package for Computing Semantic Distance in Structured Texts and Visualizing Clustering Properties in Unstructured Word Lists

Jamie Reilly[1, 2], **Emily B. Myers**[3, 4], **Hannah Mechtenberg**[4], and **Jonathan E. Peelle**[5, 6, 7]

1 Department of Communication Sciences and Disorders, Temple University, United States **2** Department of Psychology and Neuroscience, Temple University, United States **3** Department of Speech, Language, and Hearing Sciences, University of Connecticut, United States **4** Department of Psychological Sciences, University of Connecticut, United States **5** Institute for Cognitive and Brain Health, Northeastern University, United States **6** Department of Communication Sciences and Disorders, Northeastern University, United States **7** Department of Psychology, Northeastern University, United States

DOI:

Software

- Review
- Repository
- Archive

Submitted:
Published:

Summary

`SemanticDistance` computes pairwise distance between units of language (e.g., word-to-word, ngram-to-word, turn-to-turn) in both structured language samples and unstructured word lists. `SemanticDistance` has cleaning and formatting options including stopword removal and lemmatization. The package computes two complementary cosine distance indices for each pairwise contrast of interest. `SemanticDistance` can also be used to examine clustering properties within unstructured word lists, generating dendrograms and simple igraph network plots.

Software Design

`SemanticDistance` was is an open-source R package designed to accommodate a wide range of user expertise (e.g., neuroscience, linguistics, computer science). The software does not leverage artificial intelligence or use any proprietary dependencies. The package instead indexes an internal lookup database containing publicly available lexical norms.

Statment of need

`SemanticDistance` is unique in its capacity to compute semantic distance metrics within running language samples (e.g., word-to-word, bigram-to-word) and to produce semantic network visualizations. The package also provides complementary (but different) semantic distance metrics (experiential vs. embedding) computed over the same stimuli. This package will support novel insights into the flow of meaning within naturalistic language samples.